Project No. 1

Programming Assignment: Sentiment Analysis using N-grams

Assignment Goals

Understand and implement n-gram models (unigram, bigram, trigram).

Apply these models to text classification for sentiment analysis.

Perform basic feature extraction and preprocessing.

Evaluate model performance using appropriate metrics.

Conduct error analysis to identify strengths and limitations of n-gram models.

Dataset

Dataset: IMDb Movie Reviews Dataset (Small Version)

Source: https://pytorch.org/text/stable/datasets.html#imdb or https://ai.stanford.edu/~amaas/data/sentiment/

Size: 25,000 labeled reviews for training, 25,000 for testing.

Labels: Positive, Negative

Format: Plain text files, one review per file, with associated sentiment label.

Tasks

Part 1: Preprocessing
- Tokenize the text (e.g., using nltk or spaCy).
- Lowercasing, punctuation removal, optional stopword removal.
- Split data into train, dev, and test sets if not already provided.

Part 2: Feature Extraction
- Build unigram, bigram, and trigram representations of the reviews.
- Represent each review as a feature vector.
- Use sklearn's CountVectorizer with ngram_range=(1,3).

Part 3: Model Training
- Train a logistic regression, a Naive Bayes and a SVM classifier using sklearn.
- Evaluate on the validation set and compare n-gram sizes.

Part 4: Evaluation
- Use accuracy, precision, recall, F1-score.
- Plot confusion matrices.
- Optionally, analyze performance on different review lengths or polarity strengths.

Part 5: Error Analysis
- Manually inspect a small number of misclassified examples.
- Discuss linguistic causes of errors (e.g., negation, sarcasm).

Expected Deliverables

1. Python code (modular and well-commented).

2. A Jupyter notebook or script that runs the entire pipeline.

3. A short report (2–3 pages) including: method description, performance analysis, error analysis, and conclusion.

Recommended Python Libraries

Data Handling & Preprocessing: pandas, numpy, nltk, spaCy

Feature Extraction: sklearn.feature_extraction.text.CountVectorizer, TfidfVectorizer

Modeling: sklearn (MultinomialNB, LogisticRegression)

Evaluation: sklearn.metrics (classification_report, confusion_matrix)

Visualization (Optional): matplotlib, seaborn